

XEVOLAC SOLUTION FOR INJECTION

Ketorolac tromethamine

Product Name

XEVOLAC 10 mg/ml SOLUTION FOR INJECTION

XEVOLAC 30mg/ml SOLUTION FOR INJECTION

Product Description

Solution, clear to pale yellow .

Xevolac Solution for Injection can be diluted in Glucose 5.0%, Sodium Chloride 0.9%, Ringer's solution and Ringer's Lactate solution.

This product is clear and stable for 120 hours at temperature below 25°C after dilution in Sodium Chloride 0.9% solution and Ringer's solution. And this product is clear and stable for 72 hours at temperature below 25°C after dilution in Glucose 5% and Ringer's Lactate solution.

Composition

XEVOLAC INJECTION 10 mg/ml :

Each ml of injection contains Ketorolac Tromethamine 10 mg.

XEVOLAC INJECTION 30 mg/ml:

Each ml of injection contains Ketorolac Tromethamine 30 mg.

Pharmacodynamics

Ketorolac tromethamine is a nonsteroidal anti-inflammatory drug. It has an activity as anti-inflammatory, analgesic and antipyretic activity. It may inhibit the synthesis of prostaglandins in tissues by inhibiting cyclooxygenase, an enzyme that catalyzes the formulation of prostaglandin precursors (endoperoxides) from arachidonic acid.

Pharmacokinetics

Ketorolac tromethamine is absorbed after intramuscular or oral doses. At physiological pH ketorolac tromethamine dissociates to form an anionic ketorolac molecule which is less hydrophilic than the tromethamine salt. The peak plasma concentration of ketorolac is reached within about 30 to 60 minutes; absorption after intramuscular injection may be slower than that after oral doses in some individuals. Ketorolac is over 99% bound to plasma proteins. It does not readily penetrate the blood-brain barrier. Ketorolac crosses the placenta and small amounts of drug are distributed into breast milk. The terminal plasma half-life is about 4 to 6 hours, but is about 6 to 7 hours in the elderly and 9 to 10 hours in patients with renal dysfunction. The major metabolic pathway is glucuronic acid conjugation; there is some *para*-hydroxylation. About 90% of a dose is excreted in urine as unchanged drug and conjugated and hydroxylated metabolites, the remainder is excreted in the faeces.

Indications

For short-term management of moderate to severe acute post-operative pain following surgical procedures associated with low risk of haemorrhage.

Recommended Dosage

Parenteral administration:

The starting dose should be 10mg with subsequent doses of 10-30mg four to six hourly as required. The lowest effective dose should be used. The total daily dose of 90mg for the nonelderly and 60mg for the elderly should not be exceeded. Maximum duration of parenteral treatment is 2 days for all age groups. In patients who have received parenteral ketorolac and are converted to oral tablets, the total combined daily dose of all forms of ketorolac should not exceed 90mg for non-elderly and 60mg for the elderly. Maximum duration of treatment for the oral formulation is 7 days.

Xevolac solution for injection are for administration by intramuscular or bolus intravenous injection. Bolus intravenous doses should be given over no less than 15 seconds. Xevolac solution for injection should not be used for epidural or spinal administration. Dosage should be adjusted according to the severity of the pain and the patient response.

Special dosage instructions:

Elderly patients: For patients over 65 years, the lower end of the dosage range is recommended. A total daily dose of 60mg should not be exceeded.

Children: Safety and efficacy in children have not been established. Therefore, Xevolac solution for injection is contraindicated for use in children under 16 years of age.

Renal impairment: Since ketorolac trometamol and its metabolites are excreted primarily by the kidney, Xevolac solution for injection is contraindicated in moderate to severe renal impairment (serum creatinine > 160 mmol/l); patients with lesser renal impairment should receive a reduced dose (not exceeding 60 mg per day IV or IM), and their renal status should be closely monitored. After assessing the risk/benefit ratio in each individual patient, the lowest effective dose for the shortest possible duration should be used.

Mode of Administration

Intra muscular or bolus intravenous.

Contraindications

Ketorolac is contraindicated in patients with:-

- A history of peptic ulceration or gastrointestinal bleeding
- A history of haemorrhagic diathesis
- A history of confirmed or suspected cerebrovascular bleeding
- Operations associated with a high risk of haemorrhage
- A history of asthma
- Moderate or severe renal impairment (serum creatinine > 160 mmol/L)
- Hypovolaemia or dehydration from any cause
- Hypersensitivity to NSAIDs or aspirin
- During pregnancy, labour, delivery or lactation
- Concomitant administration with other NSAIDs, anticoagulant including low dose heparin

Warnings and Precautions

Ketorolac: Epidemiological evidence suggests that ketorolac may be associated with a high risk of serious gastrointestinal toxicity, relative to some other NSAIDs, especially when used outside the licensed indications and/ or for prolonged periods.

Physicians should be aware that in some patients pain relief may not occur until upwards of 30 minutes after IV or IM administration.

The use of ketorolac with concomitant NSAIDs including cyclooxygenase-2 selective inhibitors should be avoided.

Undesirable effects may be minimised by using the minimum effective dose for the shortest duration necessary to control symptoms.

Gastrointestinal ulceration, bleeding and perforation

GI bleeding, ulceration or perforation, which can be fatal, has been reported with all NSAIDs including ketorolac therapy, at anytime during treatment, with or without warning symptoms or a previous history of serious GI events.

Increased rates of clinically serious GI bleeding were seen in patients < 65 years of age who received an average daily dose of > 90 mg ketorolac IM as compared to those patients receiving parenteral opioids.

The elderly have an increased frequency of adverse reactions to NSAIDs, especially gastrointestinal bleeding and perforation which may be fatal.

The risk of GI bleeding, ulceration or perforation is higher with increasing NSAID doses, including ketorolac IV, in patients with a history of ulcer, particularly if complicated with haemorrhage or perforation, and in the elderly. The risk of clinically serious gastrointestinal bleeding is dose dependent. These patients should commence treatment on the lowest dose available. Combination therapy with protective agents (e.g. misoprostol or proton pump inhibitors) should be considered for these patients, and also for patients requiring concomitant low dose aspirin, or other drugs likely to increase gastrointestinal risk. This age-related risk of gastrointestinal bleeding and perforation is common to all NSAIDs. Compared to young adults, the elderly have an increased plasma half-life and reduced plasma clearance of ketorolac. A longer dosing interval is advisable.

NSAIDs should be given with care to patients with a history of inflammatory bowel disease, (ulcerative colitis, Crohn's disease) as these conditions may be exacerbated. Patients with a history of GI toxicity, particularly when elderly, should report any unusual abdominal symptoms (especially GI bleeding) particularly in the initial stages of treatment. When GI bleeding or ulceration occurs in patients receiving ketorolac IV, treatment should be withdrawn.

Caution should be advised in patients receiving concomitant medications which could increase the risk of ulceration or bleeding, such as oral corticosteroids, selective serotonin-reuptake inhibitors or anti-platelet agents such as aspirin.

Use in patients taking anti-coagulants such as warfarin is contraindicated.

As with other NSAIDs the incidence and severity of gastrointestinal complications may increase with increasing dose and duration of treatment with ketorolac IV. The risk of clinically serious gastrointestinal bleeding is dose-dependent. This is particularly true in elderly patients who receive an average daily dose greater than 60 mg/day of ketorolac IV. A history of peptic ulcer disease increases the possibility of developing serious gastrointestinal complications during ketorolac therapy.

Haematological effects

Patients with coagulation disorders should not receive Ketorolac. Patients on anti-coagulation therapy may be at increased risk of bleeding if given Ketorolac concurrently. The concomitant use of ketorolac and prophylactic low dose heparin (2500 - 5000 units 12-hourly) and dextrans has not been studied extensively and may also be associated with an increased risk of bleeding. Patients already on anti-coagulants or who require low-dose heparin should not receive ketorolac. Patients who are receiving other drug therapy that interferes with haemostasis should be carefully observed if Ketorolac is administered. In controlled clinical studies, the incidence of clinically significant postoperative bleeding was less than 1%.

Ketorolac inhibits platelet aggregation and prolongs bleeding time. In patients with normal bleeding function, bleeding times were raised, but not outside the normal range of two to eleven minutes. Unlike the prolonged effects from aspirin, platelet function returns to normal within 24 to 48 hours after ketorolac is discontinued.

Postoperative wound haemorrhage has been reported in association with the peri-operative use of Ketorolac IM/IV. Therefore, ketorolac should not be used in patients who have had operations with a high risk of haemorrhage or incomplete haemostasis. Caution should be used where strict haemostasis is critical, e.g. in cosmetic or day-case surgery, resection of the prostate or tonsillectomy. Haematomas and other signs of wound haemorrhage and epistaxis have been reported with the use of Ketorolac. Physicians should be aware of the pharmacological similarity of ketorolac to other non-steroidal anti-inflammatory drugs that inhibit cyclooxygenase and the risk of bleeding, particularly in the elderly.

Skin reactions

Serious skin reactions, some of them fatal, including exfoliative dermatitis, Stevens-Johnson syndrome and toxic epidermal necrolysis, have been reported very rarely in association with the use of NSAIDs. Patients appear to be at highest risk for these reactions early in the course of therapy: the onset of the reactions occurring in the majority of cases within the first month of treatment. Ketorolac should be discontinued at the first appearance of skin rash, mucosal lesions or any other sign of hypersensitivity.

SLE and mixed connective tissue disease

In patients with systemic lupus erythematosus (SLE) and mixed connective tissue disorders there may be an increased risk of aseptic meningitis.

Sodium/fluid retention in cardiovascular conditions and peripheral oedema

Caution is required in patients with a history of hypertension and /or heart failure as fluid retention and oedema have been reported in association with NSAID therapy.

Fluid retention, hypertension and peripheral oedema has been observed in some patients taking NSAIDs including ketorolac and it should therefore be used with caution in patients with cardiac decompensation, hypertension or similar conditions.

Cardiovascular and cerebrovascular effects

Appropriate monitoring and advice are required for patients with a history of hypertension and/or mild to moderate congestive heart failure as fluid retention and oedema have been reported in association with NSAID therapy.

Clinical trial and epidemiological data suggest that use of coxibs and some NSAIDs (particularly at high doses) may be associated with a small increased risk of arterial thrombotic events (for example myocardial infarction or stroke). Although ketorolac has not been shown to increase thrombotic events such as myocardial infarction, there are insufficient data to exclude such a risk for ketorolac.

Patients with uncontrolled hypertension, congestive heart failure, established ischaemic heart disease, peripheral arterial disease and/or cerebrovascular disease should only be treated with ketorolac after careful consideration. Similar consideration should be made before initiating treatment of patients with risk factors for cardiovascular disease (e.g. hypertension, hyperlipidaemia, diabetes mellitus and smoking).

Cardiovascular, Renal and Hepatic Impairment

Caution should be observed in patients with conditions leading to a reduction in blood volume and/or renal blood flow, where renal prostaglandins have a supportive role in the maintenance of renal perfusion. In these patients, administration of an NSAID may cause a dose-dependent reduction in renal prostaglandin formation and may precipitate overt renal failure. Patients at greatest risk of this reaction are those who are volume depleted because of blood loss or severe dehydration, patients with impaired renal function, heart failure, liver dysfunction, the elderly and those taking diuretics. Renal function should be monitored in these patients. Discontinuation of NSAID therapy is typically followed by recovery to the pre-treatment state. Inadequate fluid/blood replacement during surgery, leading to hypovolaemia, may lead to renal dysfunction which could be exacerbated when Ketorolac is administered. Therefore, volume depletion should be corrected and close monitoring of serum urea and creatinine and urine output is recommended until the patient is normovolaemic. In patients on renal dialysis, ketorolac clearance was reduced to approximately half the normal rate and terminal half-life increased approximately three-fold.

Renal effects

As with other NSAIDs ketorolac should be used with caution in patients with impaired renal function or a history of kidney disease because it is a potent inhibitor of prostaglandin synthesis. Caution should be observed as renal toxicity has been seen with ketorolac and other NSAIDs in patients with conditions leading to a reduction in blood

volume and/or renal blood flow where renal prostaglandins have a supportive role in the maintenance of renal perfusion.

In these patients administration of ketorolac or other NSAIDs may cause a dose-dependent reduction in renal prostaglandin formation and may precipitate overt renal decompensation or failure. Patients at greatest risk of this reaction are those with impaired renal function, hypovolaemia, heart failure, liver dysfunction, those taking diuretics and the elderly. Discontinuation of ketorolac or other non-steroidal anti-inflammatory therapy is usually followed by recovery to the pre-treatment state.

As with other drugs that inhibit prostaglandin synthesis, elevations of serum urea, creatinine and potassium have been reported with ketorolac trometamol and may occur after one dose.

Patients with impaired renal function: Since ketorolac trometamol and its metabolites are excreted primarily by the kidney, patients with moderate to severe impairment of renal function (serum creatinine greater than 160 micromol/l) should not receive Ketorolac. Patients with lesser renal impairment should receive a reduced dose of ketorolac (not exceeding 60 mg/day IM or IV) and their renal status should be closely monitored.

Use in patients with impaired liver function: Patients with impaired hepatic function from cirrhosis do not have any clinically important changes in ketorolac clearance or terminal half-life.

Borderline elevations of one or more liver function tests may occur. These abnormalities may be transient, may remain unchanged, or may progress with continued therapy. Meaningful elevations (greater than 3 times normal) of serum glutamate pyruvate transaminase (SGPT/ALT) or serum glutamate oxaloacetate transaminase (SGOT/AST) occurred in controlled clinical trials in less than 1% of patients. If clinical signs and symptoms consistent with liver disease develop, or if systemic manifestations occur, Ketorolac should be discontinued.

Anaphylactic (anaphylactoid) reactions

Anaphylactic (anaphylactoid) reactions (including, but not limited to, anaphylaxis, bronchospasm, flushing, rash, hypotension, laryngeal oedema and angioedema) may occur in patients with or without a history of hypersensitivity to aspirin, other NSAIDs or ketorolac IV. These may also occur in individuals with a history of angioedema, bronchospastic reactivity (e.g. asthma) and nasal polyps. Anaphylactoid reactions, like anaphylaxis, may have a fatal outcome. Therefore, ketorolac should not be used in patients with a history of asthma and in patients with the complete or partial syndrome of nasal polyps, angioedema and bronchospasm.

Precautions related to fertility

The use of ketorolac as with any drug known to inhibit cyclooxygenase/prostaglandin synthesis, may impair fertility and is not recommended in women attempting to conceive. In women who have difficulty conceiving or are undergoing investigation of fertility, withdrawal of ketorolac should be considered.

Fluid retention and oedema

Fluid retention, hypertension and oedema have been reported with the use of ketorolac and it should therefore be used with caution in patients with cardiac decompensation, hypertension or similar conditions.

Caution is advised when methotrexate is administered concurrently since some prostaglandin synthesis-inhibiting drugs have been reported to reduce the clearance of methotrexate, and thus possibly enhance its toxicity.

Drug Abuse and Dependence

Ketorolac is devoid of addictive potential. No withdrawal symptoms have been observed following abrupt discontinuation of ketorolac IV.

Effects on ability to drive and use machines

Some patients may experience dizziness, drowsiness, fatigue, visual disturbances, headaches, vertigo, insomnia or depression with the use of ketorolac. If patients experience these, or other similar undesirable effects, patients should not drive or operate machinery.

Interactions with other Medicaments

Ketorolac is highly bound to human plasma protein (mean 99.2%) and binding is concentration-independent.

The following medicinal products are NOT to be co-administered with Ketorolac :

Ketorolac should not be used with other ASA or other NSAIDs including cyclooxygenase-2 selective inhibitors as the risk of inducing serious NSAID-related adverse events may be increased.

Ketorolac inhibits platelet aggregation, reduces thromboxane concentrations and prolongs bleeding time. Unlike the prolonged effects from aspirin, platelet function returns to normal within 24-48 hours after ketorolac is discontinued.

Ketorolac is contraindicated in combination with anti-coagulants, such as warfarin since co-administration of NSAIDs and anti-coagulants may cause an enhanced anti-coagulant effect.

Although studies do not indicate a significant interaction between ketorolac and warfarin or heparin the concurrent use of ketorolac and therapy that affects haemostasis, including therapeutic doses of anti-coagulation therapy (warfarin) prophylactic low-dose heparin (2500-5000 units 12-hourly) and dextrans may be associated with an increased risk of bleeding.

Inhibition of renal lithium clearance, leading to an increase in plasma lithium concentration, has been reported with some prostaglandin synthesis-inhibiting drugs. Cases of increased lithium plasma concentrations during ketorolac therapy have been reported.

Probenecid should not be administered concurrently with ketorolac because of increases in ketorolac plasma concentrations and half-life.

NSAIDs should not be used for eight to twelve days after mifepristone administration as NSAIDs can reduce the effects of mifepristone.

When ketorolac is administered concurrently with oxpentifylline, there is an increased tendency to bleeding

The following medicinal products in combination with Ketorolac, are to be co-administered with caution:

As with all NSAIDs, caution should be taken when co-administering with corticosteroids because of the increased risk of gastrointestinal ulceration or bleeding.

There is an increased risk of gastrointestinal bleeding when anti-platelet agents and selective serotonin reuptake inhibitors (SSRIs) are combined with NSAIDs.

Some prostaglandin synthesis-inhibiting drugs have been reported to reduce the clearance of methotrexate, and thus possibly enhance its toxicity.

Ketorolac tromethamine does not alter digoxin protein binding. *In vitro* studies indicated that at therapeutic concentrations of salicylate (300µg/ml), the binding of ketorolac was reduced from approximately 99.2% to 97.5% representing a potential twofold increase in unbound ketorolac plasma concentrations. Therapeutic concentrations of digoxin, warfarin, ibuprofen, naproxen, piroxicam, acetaminophen, phenytoin and tolbutamide did not alter ketorolac protein binding.

Ketorolac injection reduced the diuretic response to furosemide in normovolemic healthy subjects by approximately 20% so particular care should be taken in patients with cardiac decompensation.

Co-administration with diuretics can lead to a reduced diuretic effect, and increase the risk of nephrotoxicity of NSAIDs.

As with all NSAIDs caution is advised when ciclosporin is co-administered because of the increased risk of nephrotoxicity.

There is a possible risk of nephrotoxicity when NSAIDs are given with tacrolimus.

NSAIDs may reduce the effect of diuretics and anti-hypertensive medicinal products. The risk of acute renal insufficiency, which is usually reversible, may be increased in some patients with compromised renal function (e.g. dehydrated patients or elderly patients) when ACE inhibitors and/or angiotensin II receptor antagonists are combined with NSAIDs. Therefore, the combination should be administered with caution, especially in the elderly. Patients should be adequately titrated and consideration should be given to monitoring renal function after initiation of concomitant therapy, and periodically thereafter.

NSAIDs may exacerbate cardiac failure, reduce GFR and increase plasma cardiac glycoside levels when co-administered with cardiac glycosides.

Ketorolac has been shown to reduce the need for concomitant opioid analgesia when it is given for the relief of postoperative pain.

Animal data indicate that NSAIDs can increase the risk of convulsions associated with quinolone antibiotics. Patients taking NSAIDs and quinolones may have an increased risk of developing convulsions.

NSAIDs given with zidovudine increase the risk of haematological toxicity. There is evidence of an increased risk of haemarthroses and haematoma in HIV (+) haemophiliacs receiving concurrent treatment with zidovudine and ibuprofen.

There is no evidence in animal or human studies that ketorolac trometamol induces or inhibits the hepatic enzymes capable of metabolising itself or other drugs. Hence ketorolac would not be expected to alter the pharmacokinetics of other drugs due to enzyme induction or inhibition mechanisms.

Incompatibilities

Xevolac should not be mixed in a small volume (e.g. in a syringe) with morphine sulphate, pethidine hydrochloride, promethazine hydrochloride or hydroxyzine hydrochloride as precipitation of ketorolac will occur.

It is compatible with normal saline, 5% dextrose, Ringer's, lactated Ringer's
Compatibility of Xevolac with other drugs is unknown.

Statement on Usage During Pregnancy and Lactation

Pregnancy:

In view of the known effects of NSAIDs on the foetal cardiovascular system (risk of closure of the ductus arteriosus) ketorolac is contraindicated during pregnancy, labour or delivery.

The safety of ketorolac during human pregnancy has not been established. There was no evidence of teratogenicity in rats or rabbits studied at maternally-toxic doses of ketorolac. Prolongation of the gestation period and/or delayed parturition were seen in the rat. Congenital abnormalities have been reported in association with NSAID administration in man, however these are low in frequency and do not follow any discernible pattern.

Inhibition of prostaglandin synthesis may adversely affect the pregnancy and/or the embryo/foetal development. Data from epidemiological studies suggest an increased risk of miscarriage and of cardiac malformation and gastroschisis after use of a prostaglandin synthesis inhibitor in early pregnancy. The absolute risk for cardiovascular malformation was increased from less than 1%, up to approximately 1.5 %. The risk is believed to increase with dose and duration of therapy. In animals, administration of a prostaglandin synthesis inhibitor has been shown to result in increased pre- and post-implantation loss and embryo-foetal lethality. In addition, increased incidences of various malformations,

including cardiovascular, have been reported in animals given a prostaglandin synthesis inhibitor during the organogenetic period.

During pregnancy, all prostaglandin synthesis inhibitors may expose the foetus to:

- cardiopulmonary toxicity (with premature closure of the ductus arteriosus and pulmonary hypertension);
- renal dysfunction, which may progress to renal failure with oligo-hydroamniosis;
- the mother and the neonate, at the end of pregnancy, to: possible prolongation of bleeding time, an anti-aggregating effect which may occur even at very low doses.
- inhibition of uterine contractions resulting in delayed or prolonged labour.

Ketorolac crosses the placenta to the extent of approximately 10%.

Labour and Delivery:

Ketorolac is contraindicated in labour and delivery because, through its prostaglandin synthesis inhibitory effect it may adversely affect foetal circulation and inhibit uterine contractions, thus increasing the risk of uterine haemorrhage.

There may be increased bleeding tendency in both mother and child

Lactation:

Ketorolac and its metabolites have been shown to pass into the foetus and milk of animals. Ketorolac has been detected in human milk at low concentrations therefore ketorolac is contra-indicated in mothers who are breast-feeding.

Adverse Effects/Undesirable Effects

Gastro-intestinal Disorders: The most commonly observed adverse events are gastrointestinal in nature. Peptic ulcers, ulcers, perforation or GI bleeding, sometimes fatal, particularly in the elderly, may occur. Nausea, vomiting, diarrhoea, constipation, dyspepsia, abdominal pain / discomfort, melaena, haematemesis, stomatitis, ulcerative stomatitis, eructation, flatulence, oesophagitis, gastrointestinal ulceration, rectal bleeding, pancreatitis, dry mouth, fullness, exacerbation of colitis and Crohn's disease have been reported following administration. Less frequently, gastritis has been observed.

Infection: meningitis aseptic. (especially in patients with existing auto-immune disorders, such as systemic lupus erythematosus, mixed connective tissue disease), with symptoms such as stiff neck, headache, nausea, vomiting, fever or disorientation.

Blood and Lymphatic System Disorders: thrombocytopenia

Additionally purpura, neutropenia, agranulocytosis, aplastic anaemia and haemolytic anaemia have been observed.

Immune System Disorders: anaphylaxis, anaphylactoid reactions, anaphylactoid reactions like anaphylaxis, may have a fatal outcome, hypersensitivity reactions such as bronchospasm flushing, rash, hypotension, laryngeal oedema.

These may also occur in individuals with a history of angioedema, bronchospastic reactivity (e.g. asthma and nasal polyps).

Metabolic and Nutrition Disorders: anorexia, hyperkalaemia, hyponatraemia.

Psychiatric Disorders: abnormal thinking, depression, insomnia, anxiety, nervousness, psychotic reactions, abnormal dreams, hallucinations, euphoria, concentration ability impaired, drowsiness.

Confusion and stimulation have been observed.

Nervous System Disorders: headache, dizziness, convulsions, paresthesia, hyperkinesias, taste abnormality.

Eye Disorders: abnormal vision, visual disturbances, optic neuritis.

Ear Disorders: tinnitus, hearing loss, vertigo.

Renal and Urinary Disorders: acute renal failure, increased urinary frequency, interstitial nephritis, nephrotic syndrome, urinary retention, oliguria, haemolytic uremic syndrome, flank pain (with or without haematuria +/- azotemia). As with other drugs that inhibit renal prostaglandin synthesis, signs of renal impairment, such as, but not limited to elevations of creatinine and potassium can occur after one dose of Ketorolac IV.

Cardiac Disorders: palpitations, bradycardia, cardiac failure.

Vascular Disorders: hypertension, hypotension, haematoma, flushing, pallor, postoperative wound haemorrhage.

Reproductive System and Breast Disorders: female infertility.

Respiratory, Thoracic and Mediastinal Disorders: asthma, dyspnoea, pulmonary oedema. Additionally epistaxis has been observed.

Hepatobiliary Disorders: hepatitis, cholestatic jaundice, liver failure.

Skin and Subcutaneous Tissue Disorders: exfoliative dermatitis, maculopapular rash, pruritus, urticaria, purpura, angioedema, sweating, bullous reactions including Stevens-Johnson syndrome and toxic epidermal necrolysis (very rare).

Additionally erythema multiforme and skin photosensitivity has been observed.

Musculoskeletal and Connective Tissue Disorders: myalgia, functional disorder.

General Disorders and Administration Site Condition: excessive thirst, asthenia, oedema, injection site reactions and pain, fever, chest pain.

Additionally, malaise, fatigue and weight gain has been observed.

Investigations: bleeding time prolonged, serum urea increased, creatinine increased, abnormal liver function tests

Overdose and Treatment

Overdose Symptoms and signs

Single overdoses of ketorolac have been variously associated with abdominal pain, nausea, vomiting, hyperventilation, peptic ulcers and/or erosive gastritis and renal dysfunction which have resolved after discontinuation of dosing.

Gastrointestinal bleeding may occur. Hypertension, acute renal failure, respiratory depression and coma may occur after the ingestion of NSAIDs but are rare.

Headache, epigastric pain, disorientation, excitation, drowsiness, dizziness, tinnitus and fainting have also been observed.

Rare cases of diarrhoea and occasional convulsions have been reported.

Anaphylactoid reactions have been reported with therapeutic ingestion of NSAIDs and may occur following an overdose.

Treatment

Patients should be managed by symptomatic and supportive care following NSAIDs overdose. There are no specific antidotes. Dialysis does not significantly clear ketorolac from the blood stream.

Within one hour of ingestion of a potentially toxic amount, activated charcoal should be considered. Alternatively, in adults, gastric lavage should be considered within one hour of ingestion of a potentially life-threatening overdose.

Good urine output should be ensured. Renal and liver function should be closely monitored. Patients should be observed for at least four hours after ingestion of potentially toxic amounts. Frequent or prolonged convulsions should be treated with intravenous diazepam. Other measures may be indicated by the patient's clinical condition.

Storage Conditions

Store at room temperature (below 30°C). Protect from light.

This product is clear and stable for 120 hours at temperature below 25°C after dilution in Sodium Chloride 0.9% solution and Ringer's solution. And this product is clear and stable for 72 hours at temperature below 25°C after dilution in Glucose 5% and Ringer's Lactate solution.

Dosage Forms and Packaging Available

XEVOLAC INJECTION 10 mg/ml: Box of 6 ampoules @ 1 ml.

XEVOLAC INJECTION 30 mg/ml: Box of 6 ampoules @ 1 ml.

Shelf-life

24 months

Registration Number**XEVOLAC 10 mg/ml SOLUTION FOR INJECTION - MALXXXXXXAZ****XEVOLAC 30mg/ml SOLUTION FOR INJECTION - MALXXXXXXAZ****Name and Address of Manufacturer**

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